

Saam Haghighat-Grami

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Active Military Secret Security Clearance

Education

- Kennesaw State University** *January 2022 – May 2022, January 2024 – Present*
Mechatronics Engineering, B.S. – Graduation December 2026 GPA: 4.00
- Marine Corps Communication Electronics School** *August 2022 – October 2023*
Ground Electronics Transmission Systems Maintainer GPA: 4.00
- Chattahoochee Technical College** *August 2020 – December 2021*
General Education (Undecided Major) GPA: 3.80

Technical Skills

Autonomy & Robotics: ROS2, Gazebo, Nav2, ORBSLAM3, MAVLink2, ArduPilot, Mission Planner, UAV/UGV integration, RTAB Map, Dynamical Systems & Control Theory, QGroundControl

Embedded & Hardware: ESP32, Arduino, Raspberry Pi 5, OpenMV, PX4, Matek H743 WLITE, LiDAR, RGB-D cameras, IMUs, ad hoc networking, Nema 23 stepper, N20 motors, MentorPi, Tello, ESP-NOW

Languages & Tools: C++, Python, MATLAB, Bash, Lua, Linux/WSL2, Github, VS Code, Arduino IDE, Onshape, Docker

Perception & ML: PyTorch, TensorFlow, YOLO, CNNs, LSTMs, OpenCV, sensor fusion, Optical FLOW

Manipulators & Automation: Denavit-Hartenberg parameters, UR5e, forward and inverse kinematics, trajectory planning

Research & Engineering Experience

- MEAP – Mass Estimation Autonomous Pathfinding** *November 2025 - Present*
Robotics Software Architect & Technical Lead – AgriSys Lab, Dr. Muhammad Hassan Tanveer
 - Overview:** Led a 5-person student team designing a multi-domain, multi-robot **UAV-UGV** search-and-rescue system that combines aerial mapping and ground-based navigation to identify obstacles, estimate their mass, and autonomously decide whether moving an obstacle would create a shorter path to the objective. The mass-estimation ML pipeline is targeting 85% mission success on obstacles up to 0.5 kg.
 - Achieved:** Architected the system framework and subsystem interfaces for rapid prototyping across the team. | Built ROS 2 nodes for DJITelloPy for telemetry/motion commands | Programmed ESP32 gripper control over BluetoothSerial. | COLMAP 3D point cloud for volume/mass estimation
- DARB – Drone Arm Robotic Backpack (Personal Project)** *May 2025 - Present*
Sole Designer and Builder
 - Overview:** Designed a backpack-mounted robotic arm that autonomously captures, stows, and relaunches a drone mid-flight via real-time closed-loop control, targeting a 30-second capture-to-relaunch cycle.
 - Achieved:** Built a custom autonomous drone stack on OpenMV H7+ for visual input closed-loop vehicle control via AprilTag-based visual state estimation, error correction, PID, and rpyt commands | Programmed an ESP32 for wireless comms, with PS5 manual control via Bluepad32 | Sent MAVLink2 RC overrides over UART to a Matek H743-WLITE, programmed via Mission Planner. | Built a robotic on Onshape with closed-loop steppers over ESP-NOW and IMU-based attitude state estimation and leveling control.
- Trajectory Forecasting for Floating Debris in Waterways** *June 2026 - August 2026*
Funded Undergraduate Researcher – Summer Undergraduate Research Program, Dr. Matt Marshall
 - Overview:** Designed water debris trajectory forecasting ML model with 6 variants combining scene and object-centric crops with LSPIV optical flow (OpenCV Farneback), individually and fused, to find optimal input for water debris tracking.
 - Achieved:** Designed probabilistic PyTorch CNN+LSTM+MLP model for 10s debris motion from 1s input | Achieved 75px mean error (4% frame width) via a masked Gaussian NLL loss for position/uncertainty | Surpassed the slip-corrected advection baseline by 55% mean error | Collected field storm footage on a Pi 5, detected debris with YOLOv26, built the full pipeline.
 - Publication:** First author, Submission to Winter Conference on Applications of Computer Vision (under review).
- SoilBus – Soil Monitoring Probe Network** *August 2025 - December 2025*
Lead Software Developer – AgriSys Lab, Dr. Muhammad Hassan Tanveer
 - Overview:** Designed an ad hoc soil-monitoring mesh network with auto-discovering probes and that any single node offload the entire network's data.
 - Achieved:** Full mesh connectivity in 15s and 5s per-node data offload | field-validated BFS-based routing on 5 physical nodes, scaling to 26 simulated | implemented 4 sensor types onto SD card via SPI | custom packet parsing for export.
 - Publication:** SoilBus: An Ad Hoc Soil Sensor Network for Precision Agriculture, DOI: 10.1109/SoutheastCon63549.2026.11476560, IEEE SoutheastCon 2026

Industry & Military Experience

- EOSYS Industrial and Manufacturing System Integrator** *May 2025 – August 2025*
COOP Marietta GA
 - Built an HMI display generator that produced 40+ P&ID screens for plant operators.
 - Reviewed and modified PLC programs across two production shops plus the PLC trainer system.
- United States Marine Corps Reserve** *August 2022 - Present*
Ground Radio Repair Specialist Marietta GA
 - Troubleshoot, maintain, and operate USMC radio systems in time-sensitive conditions; active Secret clearance.
 - Squad Leader of 8 Marines in 4th Combat Readiness Regiment, responsible for training, accountability, and readiness.